# Khmer News Classification Using Traditional Machine Learning Algorithms

*A Research Project Proposal for the Course Fundamentals of Machine Learning*

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## Abstract

This proposal outlines a supervised machine learning project for the automatic categorization of Khmer-language news articles. Cambodian news organizations publish large volumes of digital content daily, yet manual assignment of category labels remains time-consuming and inconsistent. The proposed study will investigate which traditional machine learning algorithm—Multinomial Naive Bayes, Logistic Regression, Linear Support Vector Machine, or Random Forest—performs best when classifying Khmer news articles, and how text preprocessing and feature representation affect classification performance. Experiments will use the KhNeAC dataset, a publicly available corpus of 7,344 labeled Khmer news articles spanning six categories, with predefined training, validation, and test splits. Text will be represented using Bag of Words and TF-IDF features, and models will be evaluated with accuracy, precision, recall, macro-averaged F1-score, and confusion matrices. The validation set will guide model selection, while held-out test performance will support the final comparison. The study is expected to identify an effective baseline approach for Khmer text classification and to yield practical insights into challenges specific to Khmer natural language processing (NLP), including the absence of whitespace word boundaries and vocabulary overlap between related categories such as politics and economics. The findings may inform future Khmer-language applications and provide a foundation for further research on low-resource NLP.

**Keywords:** Khmer natural language processing; text classification; supervised machine learning; TF-IDF; news categorization; low-resource languages

## 1. Introduction

The rapid growth of digital media has made online news platforms one of the primary sources of information in Cambodia, with major Khmer-language websites publishing large numbers of articles each day across topics such as politics, economics, entertainment, sports, technology, and daily life. Automatic categorization of this content supports search, recommendation, and archival management, yet many local platforms still rely on manual labeling, which is slow and prone to inconsistency. Supervised machine learning offers a practical alternative: a model can learn category patterns from previously labeled articles and predict the category of unseen articles automatically [3].

This proposal presents a course project for Fundamentals of Machine Learning that develops and evaluates a Khmer-language news classification system using traditional machine learning algorithms. Rather than simply building a predictive program, the project is designed as an experimental study that systematically compares multiple algorithms and feature representations, following the complete machine learning workflow from data collection to interpretation of results. Because the Khmer language presents distinctive preprocessing challenges, the project also offers an opportunity to contribute observations relevant to low-resource natural language processing.

## 2. Problem Statement

Khmer news articles are typically written as continuous text in which words are not separated by whitespace, unlike space-delimited languages such as English. This characteristic complicates tokenization, word segmentation, and feature extraction, and it may substantially affect downstream classification performance. In parallel, the practical problem remains pressing: manually assigning category labels to large article volumes consumes editorial time, and several categories share overlapping vocabulary, which increases the risk of misclassification.

The study therefore addresses the following research question: *Which traditional machine learning algorithm performs best when classifying Khmer news articles, and how do Khmer text preprocessing and feature representation affect its performance?* Answering this question requires a controlled comparison of algorithms, feature extraction methods, and segmentation strategies on a common labeled dataset.

## 3. Objectives

The project pursues five objectives: (1) collect and understand an existing, publicly available Khmer news classification dataset; (2) preprocess Khmer text and extract meaningful numerical features; (3) train multiple supervised classification algorithms under identical conditions; (4) compare model performance using appropriate evaluation metrics;

and (5) identify an effective approach for Khmer news classification and interpret the findings in the context of Khmer NLP. Together, these objectives cover every stage of the standard machine learning workflow and ensure that the study produces interpretable evidence rather than a single unexamined accuracy figure.

## 4. Related Work

Text classification is a well-established application of supervised learning in which documents are assigned to predefined categories based on their content. Classical representations convert text into numerical vectors, most commonly through Bag of Words (BoW), which counts term occurrences, and Term Frequency–Inverse Document Frequency (TF-IDF), which down-weights terms that appear across many documents [3]. Word-level and character-level n-grams extend both representations by capturing local word order and subword patterns.

Among traditional algorithms, Multinomial Naive Bayes has long served as an efficient and competitive baseline for text categorization [4], while linear Support Vector Machines have reported state-of-the-art results in classical text categorization benchmarks [5]. Logistic Regression provides a well-calibrated linear alternative, and Random Forests represent an ensemble approach. For Khmer specifically, published NLP resources remain limited, which motivates empirical studies on standard datasets. This project builds on the KhNeAC dataset released by the Cambodian Academy of Digital Technology (CADT), which provides labeled news articles together with multiple preprocessing variants [1]. All experiments will be implemented in Python using the scikit-learn library [2].

## 5. Dataset

The study will use the KhNeAC (Khmer News Article Classification) dataset published on Hugging Face [1]. The corpus contains 7,344 labeled articles divided into predefined training (5,140), validation (661), and test (1,543) partitions, enabling supervised learning and a reproducible evaluation protocol without additional data splitting. The use of the official splits also keeps model tuning statistically independent of final reporting. Table 1 summarizes the dataset; the six target categories are Economic, Entertainment, Life, Politics, Sports, and Technology. A notable property of the dataset is that it provides several preprocessing configurations, including raw, normalized, compound-word segmented, morpheme-segmented, and KCC-segmented text [1], which allows the team to examine empirically whether Khmer word segmentation improves classification—one of the central hypotheses of this study.

**Table 1. Summary of the KhNeAC dataset [1].**

| Attribute | Description |
|---|---|
| Total articles | 7,344 |
| Training / Validation / Test | 5,140 / 661 / 1,543 |
| Number of categories | 6 |
| Categories | Economic, Entertainment, Life, Politics, Sports, Technology |
| Language | Khmer |

## 6. Methodology

The project will follow a six-stage machine learning workflow: (1) dataset collection and exploratory analysis; (2) data preprocessing, including normalization, cleaning, and segmentation; (3) feature extraction using BoW and TF-IDF; (4) model training; (5) evaluation; and (6) interpretation of results and insights. The pipeline is summarized below.

Dataset Collection → Preprocessing → Feature Extraction → Model Training → Evaluation → Insights

Exploratory analysis will first examine category distribution, article length, and vocabulary size to detect class imbalance or noisy samples. Preprocessing will include Unicode normalization, removal of HTML artifacts and duplicates, and comparison of the dataset's segmentation variants. Feature extraction will convert the prepared text into document–term matrices, with vocabulary size and n-gram range treated as tunable parameters selected on the validation set.

## 7. Models and Experimental Design

Four traditional algorithms will be compared (Table 2): Multinomial Naive Bayes as a simple probabilistic baseline; Logistic Regression as a strong linear model; Linear SVM, which is typically effective for high-dimensional sparse text features; and Random Forest as an ensemble tree-based comparator. Each algorithm will first be trained with default hyperparameters, and the most promising candidates will then be tuned on the validation set. Two feature representations, BoW and TF-IDF, will be crossed with the algorithms to produce the five planned experiments listed in Table 3, allowing both the algorithms and the feature extraction methods to be compared. In addition, a supplementary experiment will evaluate the dataset's segmentation variants under the best-performing configuration to assess whether word segmentation improves classification.

**Table 2. Candidate models and their roles in the comparison.**

| Model | Role in the study |
|---|---|
| Multinomial Naive Bayes | Simple probabilistic baseline for text classification |

| Model | Role in the study |
|---|---|
| Logistic Regression | Strong linear classification baseline |
| Linear SVM | Typically effective for high-dimensional text features |
| Random Forest | Ensemble tree-based comparator |

**Table 3. Planned experiments combining features and models.**

| Experiment | Feature representation | Model |
|---|---|---|
| 1 | Bag of Words | Naive Bayes |
| 2 | TF-IDF | Naive Bayes |
| 3 | TF-IDF | Logistic Regression |
| 4 | TF-IDF | Linear SVM |
| 5 | TF-IDF | Random Forest |

## 8. Evaluation Plan

Performance will be measured with accuracy, per-class precision and recall, F1-score, macro-averaged F1-score, and confusion matrices (Table 4). Macro F1 is proposed as the primary model-selection criterion because it weights all six categories equally and prevents larger categories from dominating the assessment. The validation set will be used for model selection and hyperparameter tuning, whereas the test set will be reserved exclusively for final reporting to avoid optimistic bias. Confusion matrices will be inspected to identify the category pairs that are most frequently confused, and per-class metrics will reveal which categories are easiest or hardest to classify.

**Table 4. Evaluation metrics and what they measure.**

| Metric | What it measures |
|---|---|
| Accuracy | Overall proportion of correct predictions |
| Precision | Share of predicted articles that belong to the correct category |
| Recall | Share of actual articles that are correctly identified |
| F1-score | Harmonic mean of precision and recall |
| Macro F1 | Unweighted mean of per-class F1 across all six categories |
| Confusion matrix | Category pairs that the model confuses |

## 9. Expected Results and Insights

Because experiments have not yet been conducted, no performance values are reported in this proposal; final results will be completed after training (Table 5). The analysis is expected to answer four questions: which algorithm achieves the highest performance,

which feature representation works best, which categories are easiest to classify, and which category pairs are most frequently confused.

**Table 5. Results template to be completed after the experiments.**

| Model | Accuracy | Macro F1 |
|---|---|---|
| Naive Bayes | To be reported | To be reported |
| Logistic Regression | To be reported | To be reported |
| Linear SVM | To be reported | To be reported |
| Random Forest | To be reported | To be reported |

Several hypotheses will guide interpretation rather than being assumed as conclusions: (1) TF-IDF may outperform BoW because it attenuates high-frequency function words; (2) word segmentation may improve classification by producing cleaner lexical units, although imperfect segmentation could offset this benefit; (3) Linear SVM may outperform Naive Bayes in high-dimensional feature spaces; (4) Politics and Economics may be confused more often than other pairs because of shared vocabulary; and (5) character-level TF-IDF may add robustness when Khmer segmentation is imperfect. These hypotheses will be evaluated against empirical evidence and discussed with reference to the structural characteristics of the Khmer language.

## 10. Project Organization

For a team of five members, responsibilities will be divided as shown in Table 6. Although individual deliverables differ, all members will review the complete workflow so that the team maintains a shared understanding of the study design and results.

**Table 6. Task division for a five-member team.**

| Member | Responsibility |
|---|---|
| 1 | Introduction, problem statement, objectives, and related work |
| 2 | Dataset collection and preprocessing |
| 3 | Feature extraction and model implementation |
| 4 | Model comparison and evaluation |
| 5 | Results, insights, visualization, and presentation |

## 11. Conclusion

This proposal describes an experimental study that applies traditional machine learning algorithms to Khmer news classification using the labeled KhNeAC dataset. By comparing multiple algorithms and feature representations under a consistent evaluation protocol, the project aims to identify an effective approach for categorizing Khmer news articles and to

generate insights into the specific challenges of Khmer text processing, particularly word segmentation and vocabulary overlap between related categories. Beyond its immediate classification goal, the study is intended to demonstrate the complete machine learning workflow—from problem identification to interpretation—and to provide a foundation for future Khmer-language NLP applications.

## References

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